

KETCHIKAN, AK, USA

WMO: 703950

Lat: 55.357N

Lon: 131.712W

Elev: 80

StdP: 14.65

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25325

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	14.7	19.6	-1.1	5.2	21.5	4.7	7.0	23.4	28.1	40.2	25.6	41.5	4.2	290	0.656

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.8	72.6	60.9	69.5	59.5	66.5	58.1	62.4	70.2	60.8	67.1	59.4	64.6	10.1	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.4	75.7	63.7	58.1	72.3	62.3	57.0	69.5	61.2	28.0	70.2	26.9	67.2	26.0	64.5	67.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.9	20.7	18.3	8.2	78.7	6.2	3.4	3.7	81.1	0.1	83.1	-3.4	85.0	-7.9	87.4
			7.0	64.4	5.7	1.8	2.9	65.7	-0.5	66.7	-3.7	67.7	-7.9	69.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.8	35.8	35.7	37.8	43.1	49.7	54.9	58.5	58.7	53.5	46.2	39.4	35.7
	DBStd	10.13	7.80	5.91	5.58	4.13	4.87	4.34	4.14	3.85	3.74	4.39	5.92	6.78
	HDD50	2399	441	399	378	210	66	3	0	0	12	129	319	442
	HDD65	7021	906	819	842	657	476	306	210	201	346	583	768	907
	CDD50	869	0	0	0	3	56	150	263	269	116	11	1	0
	CDD65	16	0	0	0	0	0	3	8	5	0	0	0	0
	CDH74	58	0	0	0	0	1	15	31	10	1	0	0	0
	CDH80	5	0	0	0	0	0	2	2	1	0	0	0	0
Wind	WSAvg	8.0	8.3	8.0	8.2	8.5	8.0	8.0	7.8	7.6	7.8	8.0	8.1	8.1
Precipitation	PrecAvg	161.4	14.1	12.7	11.2	11.4	9.1	7.8	7.5	10.5	13.4	23.6	17.0	15.4
	PrecMax	185.5	31.3	23.7	24.8	22.5	21.0	14.3	17.6	21.3	27.0	42.4	43.0	24.0
	PrecMin	99.2	3.7	0.8	2.8	2.0	3.1	1.8	0.9	0.9	1.8	12.8	2.8	6.7
	PrecStd	21.9	6.4	6.2	5.6	5.4	4.5	3.1	4.5	5.5	6.4	7.8	8.6	5.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.3	50.4	55.3	62.2	70.7	76.2	78.0	75.2	67.5	58.4	52.5	50.0
		MCWB	45.1	43.9	45.9	47.7	53.9	60.8	64.2	62.8	57.9	52.1	47.2	44.5
	2%	DB	47.8	47.5	49.5	56.2	65.3	70.0	72.8	71.7	63.4	55.5	50.4	47.6
		MCWB	45.0	43.0	43.9	47.2	53.7	58.5	61.7	61.6	56.2	50.7	47.9	44.7
	5%	DB	46.0	45.4	46.9	52.6	61.6	66.1	69.2	68.7	61.2	53.9	48.7	46.2
		MCWB	43.9	42.5	43.1	45.3	52.2	56.7	60.0	60.3	55.6	50.2	46.5	43.9
	10%	DB	44.7	43.7	45.2	50.1	58.3	62.8	65.9	65.8	59.1	52.3	47.1	44.6
		MCWB	42.9	41.3	42.0	44.3	50.6	55.3	58.5	58.9	54.8	49.5	45.2	42.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.7	46.5	47.6	50.8	57.0	61.8	64.9	64.4	60.3	55.1	50.7	47.3
		MCDB	49.3	47.9	51.2	56.5	65.8	72.5	76.1	71.8	64.1	56.6	51.4	48.1
	2%	WB	46.0	44.4	45.8	48.4	54.8	59.4	62.5	62.6	58.3	53.0	48.4	46.0
		MCDB	47.2	45.8	48.5	53.0	62.6	67.1	71.2	69.4	61.3	54.1	49.6	47.1
	5%	WB	44.4	43.2	44.2	46.7	53.1	57.7	60.6	61.2	56.9	51.4	47.0	44.4
		MCDB	45.6	44.8	46.4	51.0	59.2	64.5	67.4	67.1	59.5	52.4	48.4	45.8
	10%	WB	43.3	41.7	42.6	45.7	51.5	56.0	59.2	59.8	55.8	50.2	45.6	42.8
		MCDB	44.5	43.6	44.8	49.4	56.3	61.6	64.6	64.4	58.2	51.7	47.1	44.4
Mean Daily Temperature Range	5% DB	MDBR	7.6	8.9	10.5	12.0	12.8	11.3	10.5	10.8	9.9	9.0	7.6	7.3
		MCDBR	7.5	9.4	12.8	17.0	19.6	19.0	17.2	16.5	13.5	10.1	7.5	7.5
		MCWBR	6.1	6.9	8.4	9.5	9.8	9.0	8.2	7.9	7.5	6.9	6.3	5.8
	5% WB	MCDBR	6.8	8.0	10.6	13.3	16.1	16.5	15.1	14.3	10.3	7.3	6.8	6.9
		MCWBR	6.1	6.7	7.9	8.2	8.9	8.1	7.7	7.2	6.4	6.3	6.3	6.0
Clear-Sky Solar Irradiance	taub	0.297	0.327	0.333	0.366	0.367	0.369	0.376	0.359	0.343	0.333	0.307	0.292	
	taud	2.363	2.356	2.429	2.346	2.418	2.451	2.466	2.538	2.563	2.488	2.343	2.261	
	Ebn at Noon	203	237	267	272	278	278	274	272	262	235	193	170	
	Edh at Noon	16	23	28	34	34	33	32	28	24	21	16	14	
All-Sky Solar Radiation	RadAvg	148	395	698	1159	1449	1504	1341	1175	764	418	186	114	
	RadStd	20	81	121	93	153	120	178	117	106	67	32	21	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-342	+151	N/A
Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+108	N/A

Nomenclature: See separate page